

PRAVEEN PAL RAWAL

AI / ML Engineer

Jodhpur, Rajasthan, India | +91 85296 23327 | rawalpraveen886@gmail.com

 linkedin.com/in/praveen-rawal |  github.com/thepietro23

Education

M.Tech, Data Science & Machine Learning

2024 – 2026 (Expected)

Rashtriya Raksha University, Gandhinagar, Gujarat | CGPA: 8.52

B.Tech, Computer Science & Engineering (AI)

2019 – 2023

Vivekananda Global University, Jaipur, Rajasthan | CGPA: 9.1 | Silver Medalist (2nd Rank)

Experience

AI/ML Intern | *Forensic Cybertech*

Jan 2026 – Present

- Built an EDR behavioral malware-detection model focused on real-time **ransomware detection**, profiling malicious process behavior rather than relying on static signatures.
- Engineered a **UEBA** (User & Entity Behavior Analytics) system using 5 ML models to model normal user behavior, detect anomalous activity, and trigger automated alerts.
- Developing **agentic AI** systems (autonomous LLM agents) to safeguard the product against emerging threats.
- Containerized and deployed models on **AWS** with Docker, using Apache Kafka for data streaming and LangChain for LLM workflows.

GenAI Backend Developer Intern | *Techity Solutions Pvt. Ltd.*

Oct 2025 – Jan 2026

- Developed AI-powered backend systems and REST APIs integrating large language models (LLMs) for production-grade GenAI applications.
- Designed and integrated LLM pipelines into backend services to deliver reliable, deployable AI features.
- Implemented and tested API endpoints for AI features with robust error handling and reliable LLM response processing.

AI Engineering Intern | *AI Core System*

Jun 2025 – Aug 2025

- Engineered an end-to-end speech-to-email automation pipeline using Whisper ASR, LangChain, and FastAPI; cut manual reporting time by ~83%.
- Added silence detection, input validation, and retry logic for noisy audio, achieving <2s end-to-end latency.
- Integrated the SendGrid API for automated, templated dispatch of 50+ daily reports with error handling.

AI Workshop Trainer | *NCII SECEX, Rashtriya Raksha University*

Apr 2025

- Delivered applied AI/ML workshops on cybersecurity use cases to 150+ professionals and government officials.
- Designed curriculum on neural networks, anomaly detection, NLP for threat intelligence, and deployment with FastAPI & Docker.

Projects

ResearchSynth AI — Multi-Agent Research Platform | *LangGraph, RAG, FastAPI*

2025

- A multi-agent platform that autonomously researches a topic and produces a structured, cited synthesis report.
- Built a 6-agent + critic system in LangGraph with conditional feedback loops and a RAG pipeline (ChromaDB + Groq) for <1.5s inference.
- Delivered a full-stack app (FastAPI + React) with real-time SSE streaming and multi-format (PDF / Markdown) export.

- ▶ A RAG advisor that recommends realistic, budget-aware AI/PC hardware builds, eliminating LLM hallucinations and over-priced suggestions.
- ▶ Combined FAISS semantic search (384-dim MiniLM) with budget/VRAM filtering to ground Llama-3.3-70B (Groq), plus a constraint-based build assembler (PSU / RAM / socket rules).
- ▶ Delivered cited, budget-constrained build recommendations with a buy-vs-rent break-even model through a Streamlit conversational UI.

Financial Sentiment LLM — QLoRA Fine-Tuning | QLoRA, PyTorch, Transformers

2025

- ▶ A fine-tuned LLM that classifies financial-market sentiment (bullish / bearish / neutral) from news headlines and tweets.
- ▶ Fine-tuned Qwen2.5-0.5B-Instruct using QLoRA (4-bit NF4 + LoRA, rank 16 / α 32) on 9,500+ samples, training only **1.75%** of parameters on a single 4 GB consumer GPU.
- ▶ Improved test accuracy from **45%** to **81.7%** on a 2,388-example set (beating the ~46% majority baseline), with bullish-sentiment recall jumping 0.06 $\rightarrow$ 0.74 and a portable 35 MB LoRA adapter.

Skills

Programming Python, SQL (MySQL, PostgreSQL), Bash

AI / ML PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Hugging Face Transformers

LLMs &LangChain, LangGraph, RAG, Agentic AI, Prompt Engineering, Fine-Tuning (QLoRA, PEFT /

GenAI LoRA, TRL), Groq, OpenAI, Ollama

NLP Whisper ASR, BERT, FinBERT, spaCy, NER, Sentence Transformers

Data & MLOps Apache Kafka, ChromaDB, FAISS, Docker, FastAPI, Streamlit, AWS, Git, GitHub Actions, CI/CD

Achievements & Leadership

- ▶ **Runner-up — Smart India Hackathon (National), 2022:** competed against 10,000+ teams nationwide.
- ▶ **Founder, AI Club — RRU:** established the university AI club; led applied ML sessions for 50–60 students and mentored juniors on training, deployment, and research.
- ▶ **Certifications:** Neural Networks & Deep Learning (DeepLearning.AI); Big Data Fundamentals (IBM) — Coursera, 2025.
- ▶ **Junior Under Officer, NCC (1 Raj Armed SQN):** commanded 120 cadets; earned the NCC ‘C’ Certificate with Grade A.
- ▶ **Liaison & Operations Coordinator, AIPSC (Nov 2024):** managed end-to-end logistics for a national conference with 200+ delegates, including senior IPS/IAS officials.